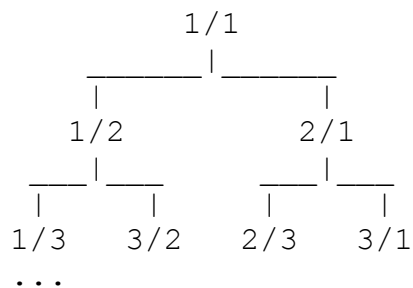


Kick Start 2013 - Round A

Analysis: Cross the maze

Kick Start 2013 - Round A

Analysis: Rational Number Tree



The path to the n -th element in the tree can be presented by the binary representation of n . For example, if $n = 6$, then its path can be represented by 110, meaning "1 (root, $1/1$) $\Rightarrow$ 1 (go right, $2/1$) $\Rightarrow$ 0 (go left, $2/3$)", and we get $2/3$. So to solve Q1, we at first find the binary representation of n , then goes down the tree along the path.

To solve Q2, we need to find the path from p/q to $1/1$. To achieve this we need to continuously subtract p from q (if $q > p$) or subtract q from p (if $p > q$). Hence, the path can be constructed by representing these two operations by 0 and 1 respectively.

Code list:

```
import fractions
```

```
def Normalize(p, q):
    r = fractions.Fraction(p, q)
    return r.numerator, r.denominator
```

```
def FindElement(n):
    """Solution for Q1."""
    assert n > 0
    path = bin(n)[3:]
    p = q = 1
    for x in path:
        if x == '0':
            q += p
        else:
            p += q
    return p, q
```

```
def FindPosition(p, q):
    """Solution for Q2."""
    assert p > 0 and q > 0
    p, q = Normalize(p, q)
    b = ''
    while p != q:
        if p > q:
            p -= q
            b = '1' + b
        else:
            q -= p
            b = '0' + b
    return b
```

```
    q -= p
    b = '0' + b
b = '1' + b
return int(b, 2)
```

Kick Start 2013 - Round A

Analysis: Read Phone Number

This is a very easy problem, the main examination point is to test the students whether they know how to split string, convert string into number.

Kick Start 2013 - Round A

Analysis: Sorting

If you know sorting, you must know how to solve this problem :)

Analysis: Spaceship Defence

For each query:

We can consider there are edges with weight 0 between every two nodes with same color. Then we can run a Dijkstra algorithm to get the distance of shortest path. In that case the time complexity would be $O(Q * N * \log N + M)$ and space complexity would be $O(N + M)$. In which N is the number of vertexes, M is the number of edges and Q is the number of query.

If we make the vertexes with same color into one single vertex, we can reduce the graph and then reduce the problem. In that case, the time complexity would be $O(Q * C * \log C + N + M)$. In which C is the number of different color.